

Problem 7.18

If

$$\frac{i}{1+i} = 0.06,$$

find i and v .

Solution

Starting from

$$\frac{i}{1+i} = 0.06,$$

multiply both sides by $(1+i)$:

$$i = 0.06(1+i).$$

Expanding,

$$i = 0.06 + 0.06i.$$

Thus,

$$i - 0.06i = 0.06,$$

so

$$0.94i = 0.06.$$

Hence,

$$i = \frac{0.06}{0.94} \approx 0.0638298.$$

Therefore,

$$\boxed{i \approx 6.3830\%}.$$

Now,

$$v = \frac{1}{1+i} = \frac{1}{1.0638298} \approx 0.94.$$

So,

$$\boxed{v = 0.94}.$$